

Most P&S cameras have a crop factor of 1.3 to 2.0. This is what marks the difference between ‘number of sensor pixels’ and ‘effective pixels’. Some of the cameras lose effective pixels because the camera’s lens can’t cover the entire sensor area and so what you get is actually a picture that’s been cropped all around. At the same time, full frame sensors like the Canon 1D, sometimes lead to a visible fringing – a lateral chromatic aberration – which you’ll need to crop with editing software after the fact. But all said and done, the full frame sensor gives you a wide-angle of view (therefore capturing more of the scene) and a better quality image.

14.5 Sensor size and image resolution

Cameras with smaller sensors use shorter focal length lenses to get the same angular coverage as cameras with larger sensors do with longer focal length lenses. So what’s the big deal about sensor size, then? The size of the sensor affects many factors in the image and the functions of the camera and its body. Basically, sensors are available in three different sizes : four thirds, APS and Full Film Format.

Sensor	Ratio	Resolution	File Size	Printed Sizes					
size	format	height x width	Mb's	6" x 4"	7" x 5"	8" x 6"	10" x 8"	A4	A3
24.5mp	3x2	6048 x 4034	74.4mb	1000px	940px	720px	680px	507px	466px
20.1mp	3x2	5616 x 3744	53.3mb	916px	775px	583px	514px	400px	372px
16.7mp	3x2	4992 x 3328	50.1mb	832px	688px	509px	452px	341px	312px
14mp	3x2	4608 x 3024	42mb	768px	637px	480px	417px	317px	275px
12mp	3x2	4308 x 2848	38mb	712px	590px	450px	392px	300px	265px
9mp	3x2	3872 x 2592	31mb	640px	530px	400px	350px	262px	234px
8mp	4x3	3264 x 2448	24mb	576px	475px	360px	318px	243px	211px
5mp	3x2	3008 x 2000	18mb	512px	411px	300px	270px	204px	177px
5mp	4x3	2880 x 1920	16mb	480px	375px	270px	240px	180px	161px
4mp	4x3	2240 x 1680	12mb	384px	300px	216px	177px	135px	111px
3mp	4x3	2048 x 1536	8mb	362px	275px	192px	156px	118px	92px

Making sense of sensor resolution